Scan2Tally

User Stories, Roles & Database Requirements

Working Product Requirements Document

1. Product Overview

Scan2Tally is a web application that converts handwritten bill and receipt images into structured transaction data and generates an Excel file in the exact format required by the client for import into Tally.

Current workflow:

- Handwritten Receipt → Manual Data Entry → Excel → Tally

Scan2Tally workflow:

- Upload Receipt → Extract Data → Review/Edit → Generate Excel → Import into Tally

The application should remove the manual typing step while keeping the client's existing Tally import workflow unchanged.

2. User Roles and Permissions

The system has three roles: Super Admin, Admin, and Operator.

Permission	Super Admin	Admin	Operator
View all organizations	Yes	No	No
Manage organizations	Yes	No	No
View all users	Yes	Organization only	No
Create/manage users	Yes	Yes, within scope	No
Change roles	Yes, except Super Admins	Users below their role	No
View all organization data	Yes	Yes	No
View own data	Yes	Yes	Yes
Create Scan Batches	Yes	Yes	Yes
Upload receipts	Yes	Yes	Yes
Extract data	Yes	Yes	Yes
Edit transaction data	Yes	Yes	Own data
Generate Excel	Yes	Yes	Yes
Edit profile	Yes	Yes	Yes
Change password	Yes	Yes	Yes

2.1 Super Admin

The Super Admin represents the Scan2Tally platform owner.

- Create, view, edit, and deactivate client organizations.
- Create and manage users.
- Assign and change user roles, except the role of another Super Admin.
- View all organizations' data.
- View all Scan Batches, receipts, and transactions.

- Monitor overall storage usage.
- Edit their own profile and change their password.

Restriction: A Super Admin cannot change or remove another Super Admin's role.

2.2 Admin

The Admin belongs to a specific client organization.

- View all Scan Batches and transaction data belonging to their organization.
- Create users within their organization.
- Edit users within their permitted scope.
- Change roles of users below them.
- Deactivate users.
- Edit their own profile and change their password.

Restrictions: An Admin cannot change their own role, modify Super Admin accounts, or access another organization's data.

2.3 Operator

The Operator is the primary user who processes receipts.

- Create a new Scan Batch.
- Rename a Scan Batch.
- View Scan Batches they created.
- Upload one or multiple receipt images.
- Start extraction manually.
- View extraction status.
- Retry failed extraction.
- Add new receipts to an existing Scan Batch.
- View original receipt images.
- View and edit extracted transaction data.
- Save progress.
- Generate and download Excel.
- Edit their own profile and change their password.

Restriction: An Operator can only view and work with Scan Batches and data they created.

3. User Stories

Epic 1: Authentication and Profile

US-01: Login

As a registered user, I want to log in with my email and password so that I can access Scan2Tally.

Acceptance criteria:

- The user can enter an email and password.
- Invalid credentials show an error.
- Successful login redirects the user to the appropriate dashboard.
- The user can only access features permitted by their role.

US-02: Logout

As a user, I want to log out so that my account cannot be accessed by someone else using my device.

US-03: Change Password

As a user, I want to change my password so that I can maintain the security of my account.

US-04: Forgot Password

As a user, I want to reset my forgotten password so that I can regain access to my account.

US-05: Edit Profile

As a user, I want to edit my profile information so that my information remains up to date.

Acceptance criteria:

- Editable profile information may include name, phone number, and designation.
- Email should preferably remain the login identifier and not be freely editable.

Epic 2: Organization Management

US-06: Create Organization

As a Super Admin, I want to create a client organization so that the client can use Scan2Tally.

US-07: Edit Organization

As a Super Admin, I want to edit organization information so that client information remains accurate.

US-08: Deactivate Organization

As a Super Admin, I want to deactivate an organization so that users belonging to an inactive client can no longer access the system.

Acceptance criteria:

- Historical data should not be automatically deleted.

US-09: Organization Data Isolation

As the system, I want every organization's data to be isolated so that one client cannot access another client's data.

Epic 3: User Management

US-10: Create User

As a Super Admin or Admin, I want to create a user so that authorized employees can access Scan2Tally.

Acceptance criteria:

- User information includes name, email, phone number, designation, role, and organization.
- The system sets or generates an initial password securely.

US-11: Assign Role

As an authorized administrator, I want to assign a role to a user so that the user receives the correct permissions.

US-12: Edit User

As an authorized administrator, I want to edit user information so that user details remain accurate.

US-13: Change User Role

As an authorized administrator, I want to change a user's role so that their system access matches their responsibilities.

Acceptance criteria:

- A user cannot change their own role.
- A user cannot promote themselves beyond their permitted scope.

US-14: Deactivate User

As an authorized administrator, I want to deactivate a user so that inactive employees cannot log in.

Acceptance criteria:

- Historical data created by the user must remain available.

Epic 4: Scan Batch Management

US-15: Create Scan Batch

As an Operator, I want to create a Scan Batch so that I can organize a group of receipt images and their extracted data together.

Acceptance criteria:

- A default name is generated, such as 'Scan Batch - July 23, 2026'.

US-16: Rename Scan Batch

As an Operator, I want to rename a Scan Batch so that I can easily identify it later.

US-17: View Scan Batches

As a user, I want to view the Scan Batches I am permitted to access so that I can continue working on them later.

US-18: Continue Existing Scan Batch

As a user, I want to open an existing Scan Batch so that I can continue working where I left off.

US-19: Save Progress

As a user, I want my Scan Batch progress to be saved so that I can leave and continue later without losing my work.

Epic 5: Receipt Management

US-20: Upload Receipt

As an Operator, I want to upload one or multiple receipt images so that I can process them.

US-21: View Uploaded Receipts

As an Operator, I want to see uploaded receipt images and their extraction status so that I know which files need processing.

Acceptance criteria:

- Supported statuses: Pending, Processing, Extracted, and Failed.

US-22: Add More Receipts

As an Operator, I want to add more receipt images to an existing Scan Batch so that I can continue adding receipts later.

US-23: Delete Receipt

As an authorized user, I want to delete an incorrectly uploaded receipt before extraction so that unwanted files are not processed.

Acceptance criteria:

- For the MVP, deletion may be limited to receipts with Pending status.

Epic 6: AI Extraction

US-24: Start Extraction

As an Operator, I want to manually start extraction so that the system can extract transaction data from my uploaded receipts.

US-25: Process Only Unprocessed Receipts

As the system, I want to process only new or failed receipts so that successfully extracted receipts are not processed again.

US-26: Handle Partial Failure

As the system, I want to save successful extraction results even if other receipts fail so that successful work is not lost.

US-27: Retry Failed Extraction

As an Operator, I want to retry a failed receipt so that I can attempt extraction again.

Epic 7: Review and Editing

US-28: Review Receipt

As an Operator, I want to view the original receipt image and extracted data side-by-side so that I can compare the extracted information with the original receipt.

US-29: Edit Extracted Data

As an Operator, I want to edit extracted transaction fields so that I can correct inaccurate information.

US-30: Save Extracted Data

As an Operator, I want to save my changes so that the corrected information is used for Excel generation.

Epic 8: Excel Export

US-31: Generate Excel

As an Operator, I want to generate an Excel file from the transaction data so that I can import it into Tally.

US-32: Generate Tally-Compatible Format

As the system, I want to generate the Excel file using the required client format so that it can be imported into Tally without changing the client's existing workflow.

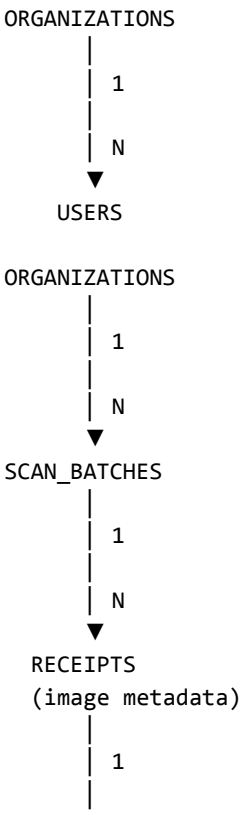
US-33: Regenerate Excel

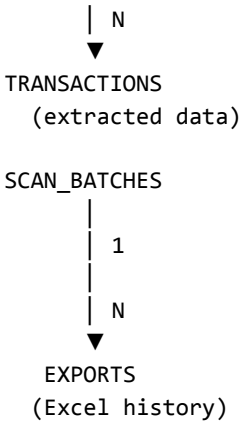
As a user, I want to generate the Excel file again from the latest saved transaction data so that I can download an updated version.

4. Database Design

The database should be designed to support the current client and future multiple-client scalability.

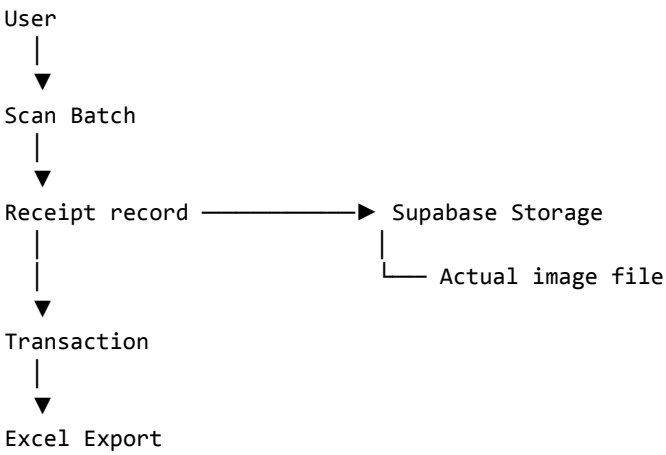
4.1 Entity Relationship Diagram





4.2 Data Flow

The actual receipt image should be stored in Supabase Storage. The database stores only metadata and the path to the image.



4.3 Table: organizations

Column	Type	Description
id	UUID	Primary key
name	Text	Client/company name
status	Enum	Active / Inactive
created_at	Timestamp	Creation time
updated_at	Timestamp	Last update

4.4 Table: users

Authentication credentials should be managed by Supabase Auth. Do not store plain-text passwords in the application database.

Column	Type	Description
id	UUID	Same user ID as Supabase Auth
organization_id	UUID / NULL	Organization the user belongs to; NULL for Super Admin
name	Text	Full name
email	Text	Login email
phone	Text	Phone number
designation	Text	Job designation

role	Enum	Super Admin / Admin / Operator
status	Enum	Active / Inactive
created_by	UUID / NULL	User who created the account
last_login_at	Timestamp / NULL	Last login
created_at	Timestamp	Creation time
updated_at	Timestamp	Last update

4.5 Table: scan_batches

Column	Type	Description
id	UUID	Primary key
organization_id	UUID	Client organization
created_by	UUID	User who created the batch
name	Text	Batch name
status	Enum	Active / Archived
created_at	Timestamp	Creation time
updated_at	Timestamp	Last modification

4.6 Table: receipts

The receipt table stores metadata only. The actual image is stored in Supabase Storage.

Column	Type	Description
id	UUID	Primary key
batch_id	UUID	Related Scan Batch
uploaded_by	UUID	User who uploaded the image
file_path	Text	Path to image in Supabase Storage
status	Enum	Pending / Processing / Extracted / Failed
error_message	Text / NULL	Extraction error, if any
created_at	Timestamp	Upload time

4.7 Table: transactions

This table stores the structured data extracted from the receipt and any corrections made by the user.

Column	Type	Description
id	UUID	Primary key
receipt_id	UUID	Related receipt
batch_id	UUID	Related Scan Batch
organization_id	UUID	Organization
[Tally fields]	Various	Fields required by the client's Tally Excel format
created_at	Timestamp	Creation time
updated_at	Timestamp	Last update
updated_by	UUID	Last user who edited the data

Important: The exact transaction columns must be based on the client's existing Tally import Excel format.

4.8 Table: exports

Column	Type	Description
id	UUID	Primary key
batch_id	UUID	Related Scan Batch
generated_by	UUID	User who generated the export
file_name	Text	Excel filename
record_count	Integer	Number of transaction records
created_at	Timestamp	Export generation time

Recommendation: Do not permanently store every Excel file. Generate the Excel file from the latest saved transaction data when the user requests an export. The exports table can retain export history.

5. Core User Flow

```
LOGIN
  ↓
DASHBOARD
  ↓
Create New Scan Batch OR Continue Existing Scan Batch
  ↓
Upload One or Multiple Receipt Images
  ↓
Click "Extract Data"
  ↓
Each receipt is processed independently
  ├── Success → Extracted
  └── Failure → Failed / Retry Available
  ↓
View Receipt Image + Extracted Data Side-by-Side
  ↓
Edit Data if Necessary
  ↓
Save Changes
  ↓
Add More Receipts Later OR Generate Excel
  ↓
Download Excel
  ↓
Import into Tally
```

6. MVP Scope

The first version should focus on the core workflow.

Must have:

- Authentication and role-based access.
- Super Admin, Admin, and Operator roles.
- Organization structure for future multi-client scalability.
- User management.
- Create and rename Scan Batches.
- Upload one or multiple receipt images.
- Manual extraction button.
- Individual extraction status for each receipt.
- Retry failed extraction.
- Add new receipts to existing Scan Batches.
- Side-by-side receipt and extracted data review.
- Edit and save extracted transaction data.
- Generate Tally-compatible Excel.

Not required for the first version:

- Complex analytics.
- AI confidence scores.
- Multi-level approval workflows.

- Advanced audit history.
- Complex Excel versioning.
- Permanent storage of every Excel file.
- Multiple advanced roles.

7. Key Development Notes

- The client's existing Tally Excel format must be treated as the source of truth for transaction fields.
- The actual receipt image should be stored in Supabase Storage, while the database stores metadata and the file path.
- Successful receipt extractions must not be processed again unless the user explicitly retries or reprocesses them.
- A failed receipt must not cause successful receipts in the same batch to be lost.
- All organization-related data must be isolated between clients.
- The system should support a future multi-client setup even though the initial deployment may have only one client.
- The user should be able to leave a Scan Batch and continue later.